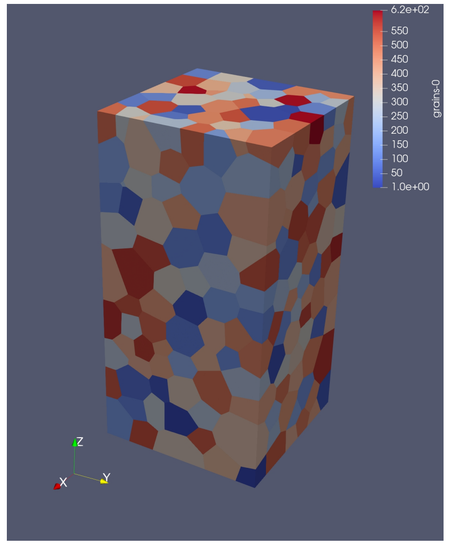
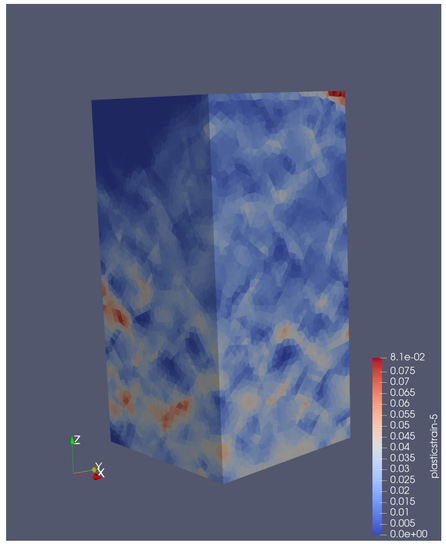
Mechanical Banding (MechBands)

Purpose

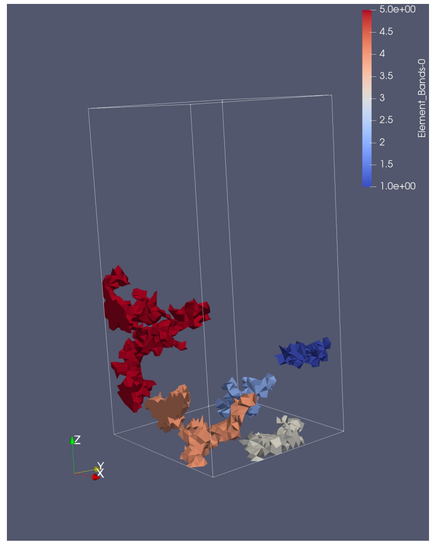
Identify sets of connected elements for:
(1) the mesh of a polycrystal and
(2) a specified range of values of a designated field variable.
(3) Sort the sets by size and save the largest ones (currently five sets are saved).
(4) Create graph objects of the larger sets for the field variable.



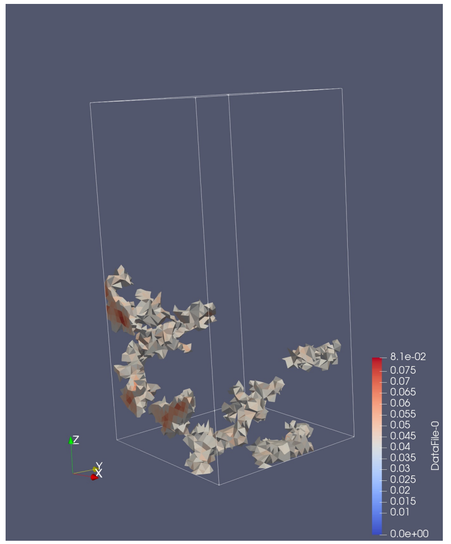
(1) Grain Structure



(2) Effective Plastic Strain



(3) Five Largest Bands



(4) Effective Plastic Strain

Mechanical Banding (MechBands)



Methodology

Pre-process mesh data:

The code first computes connectivity arrays that give the neighboring elements for every element in the mesh. From this, the code computes a mesh link array: for every interface between two elements, the numbers of the elements sharing the interface. This step is time-consuming, but needs to be done only once for a mesh. The result is saved to a file for re-use.

Pre-process field data:

The code next determines if the field variable value for each element lies within a specified range of values and assigns a T/F status.

Construction of connected sets:

The code then steps through all of the elements of the mesh to determine if an element should be part of an existing connected set.

1. If so, the element is added to the set.
2. If it should be part of more than one set, the code adds the element to both sets and merges the sets.
3. If the element satisfies the condition that it could be in a set but no connected set exists for it, the code creates a new set and inserts the element in that set.

Once all elements have been examined, the code sorts the sets by size in descending order and then defines the sets as bands.

Methodology (continued)

Auxiliary construction:

Using a MatLab routine, the code also creates undirected graphs for the bands wherein the graph nodes are finite elements and the graph links are element interfaces.

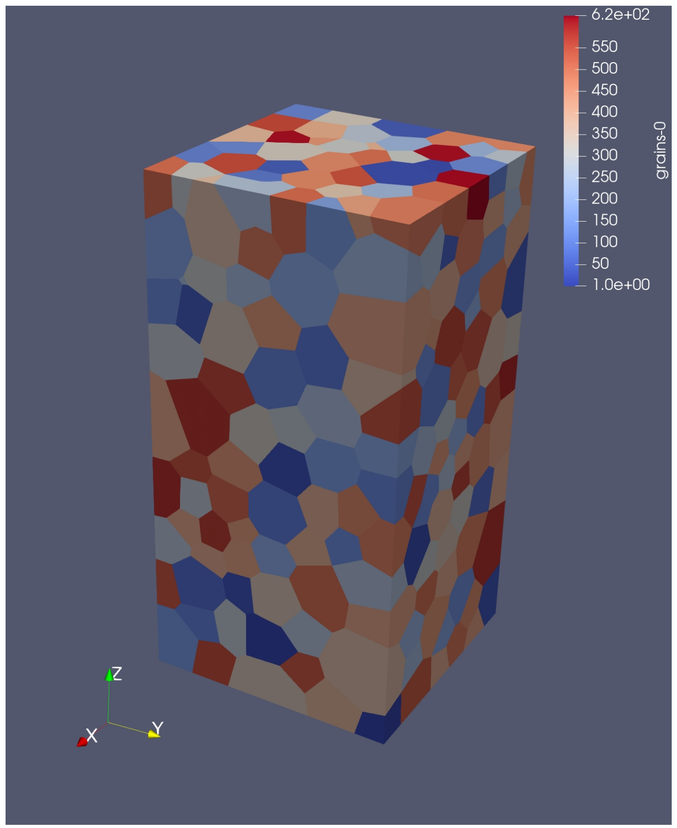
The band graphs are related to a graph of the mesh (again graph nodes are finite elements and the graph links are element interfaces).

Finally, the code identifies links of the mesh graph for which both connected elements are within the data range and creates a new graph consisting only of these links.

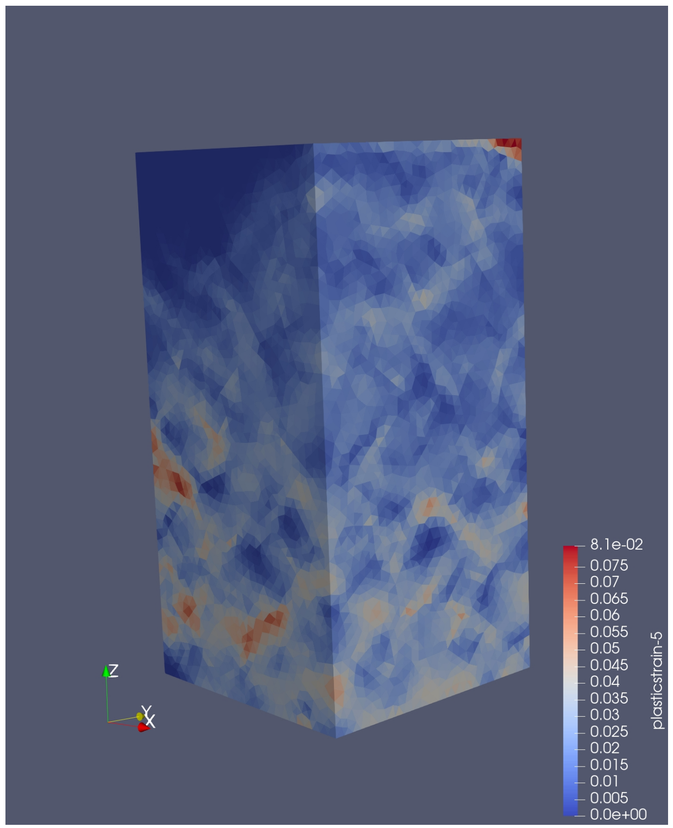
Mechanical Banding (MechBands)

Example 1: Formation of Deformation Bands within a virtual polycrystal.
Boundary conditions: Z Extension with gripped upper boundary

Grain Structure



Effective Plastic Strain

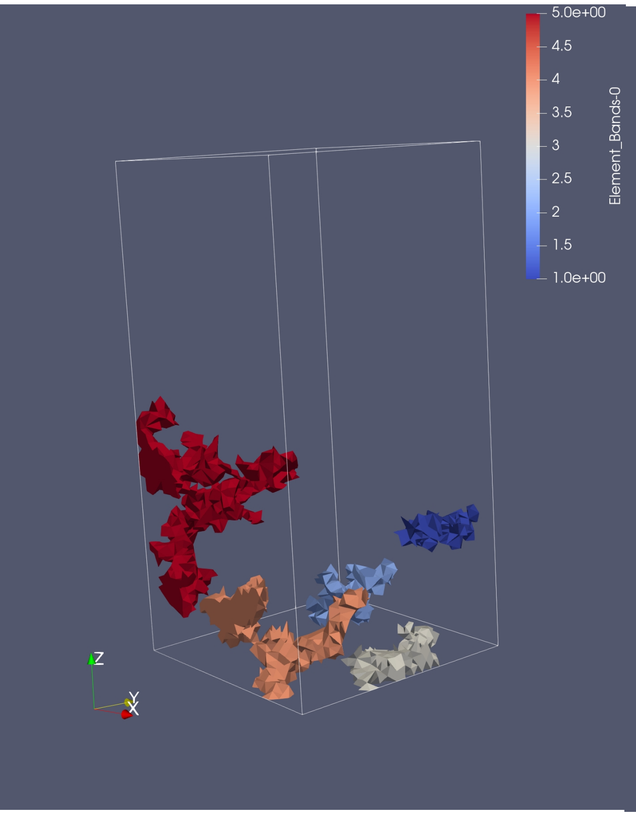


Field variable: plastic strain distribution
Deformation state: 2% nominal strain
Solution generated by FEpX.
Polycrystal has 624 grains.
Mesh has ~225,000 elements.

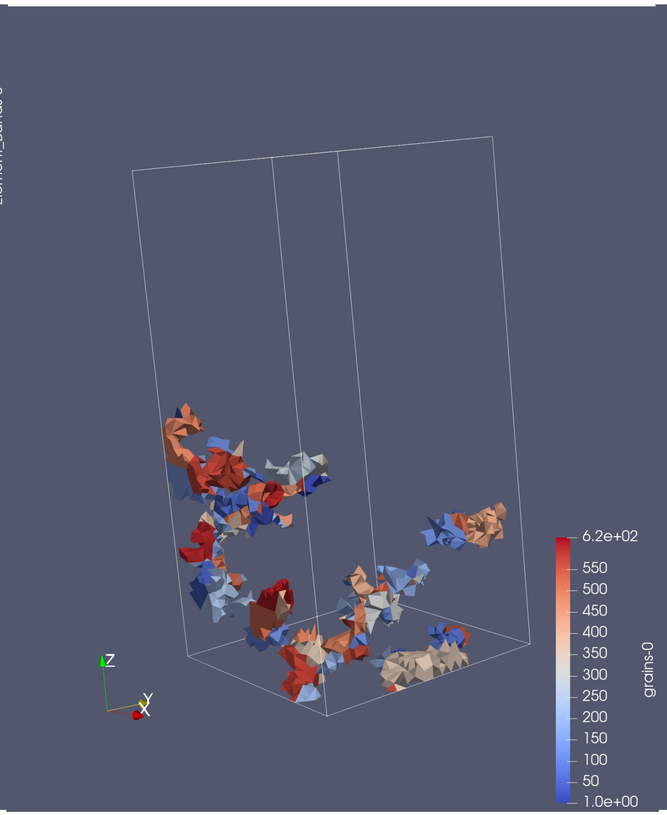
Mechanical Banding (MechBands)

Example1: Formation of Deformation Bands within a virtual polycrystal.
Boundary conditions: Z Extension with gripped upper boundary

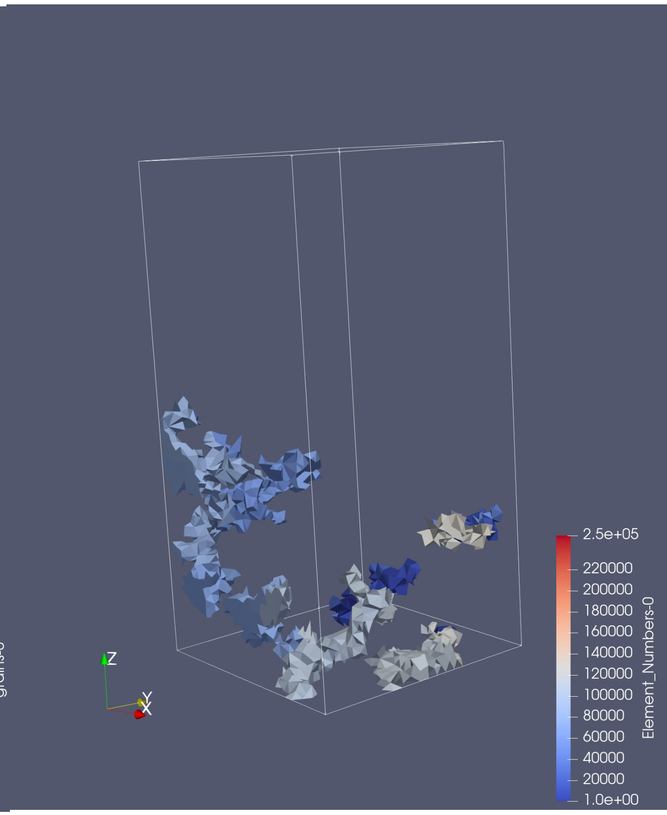
Five Largest Bands



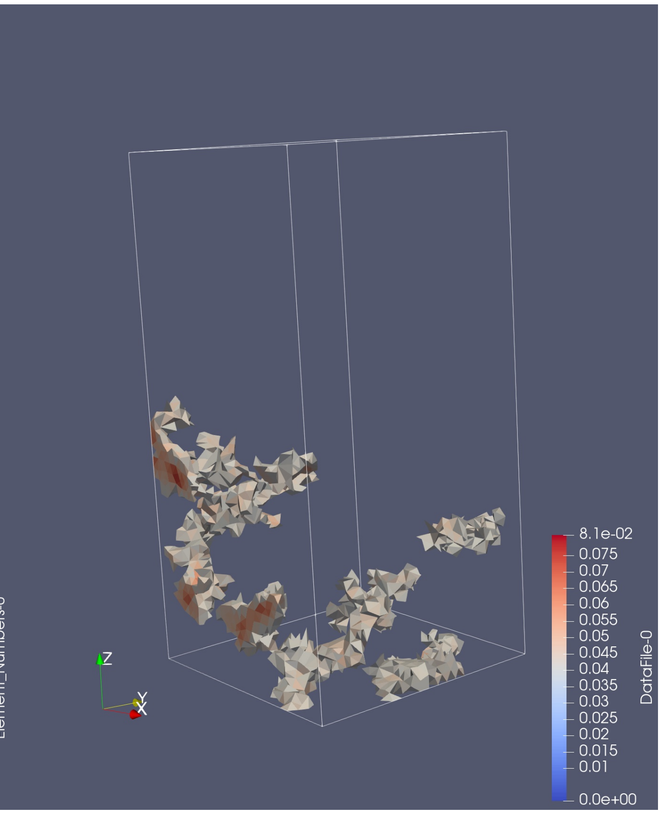
Grain Numbers within Bands



Element Numbers within Bands



Effective Plastic Strain

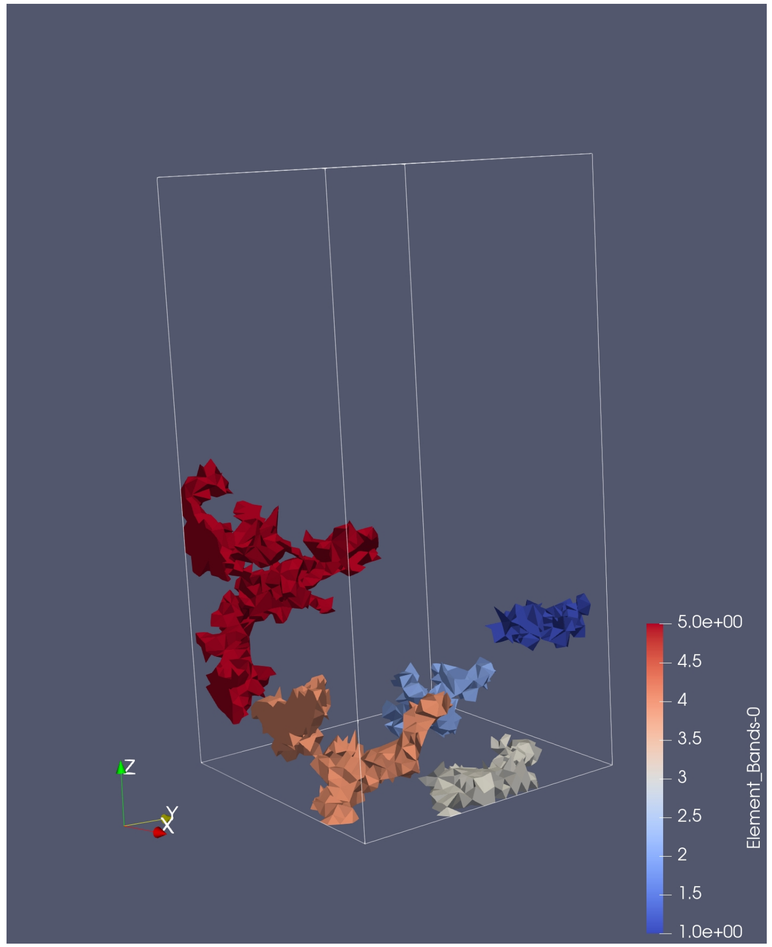


Threshold plastic strain range: 0.04 to 0.09
(nominal strain of 0.02)

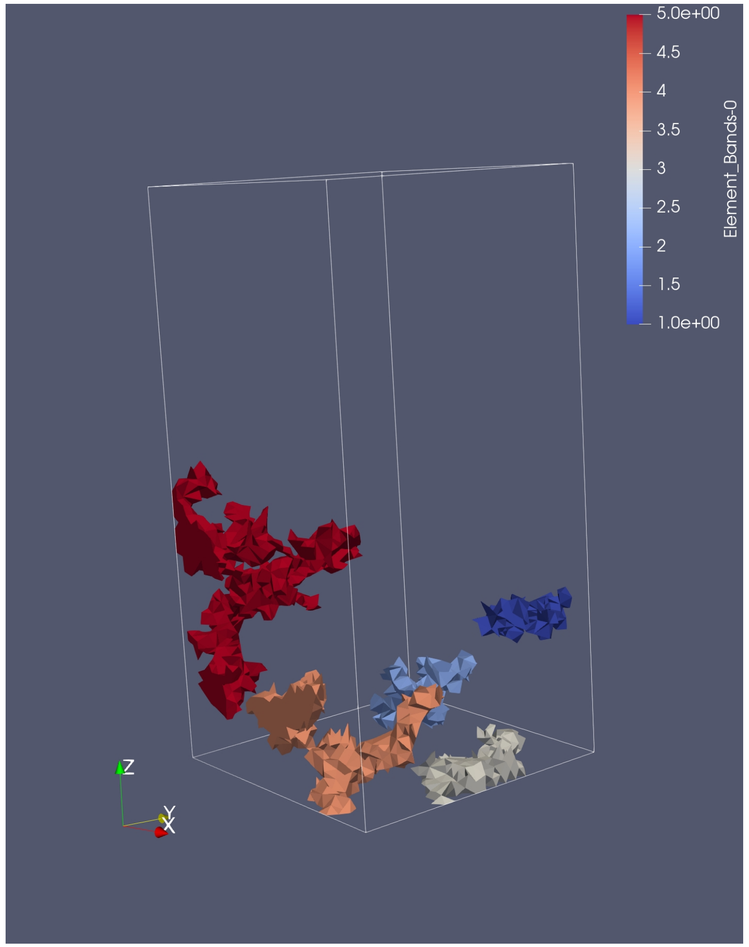
Mechanical Banding (MechBands)

Example 1: Formation of Deformation Bands within a virtual polycrystal.
Boundary conditions: Z Extension with gripped upper boundary

MatLab 'conncomp' graph routine



MechBands search method

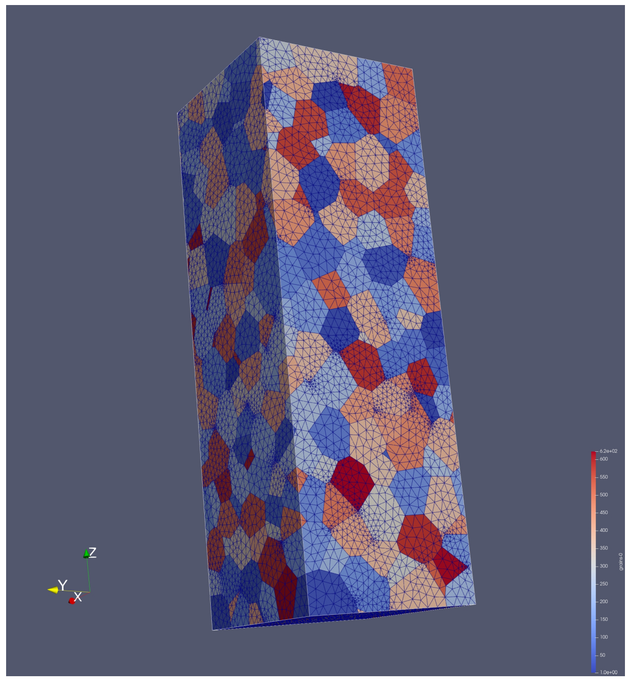


Two methods identify the same sets of connected elements for the 5 largest bands.

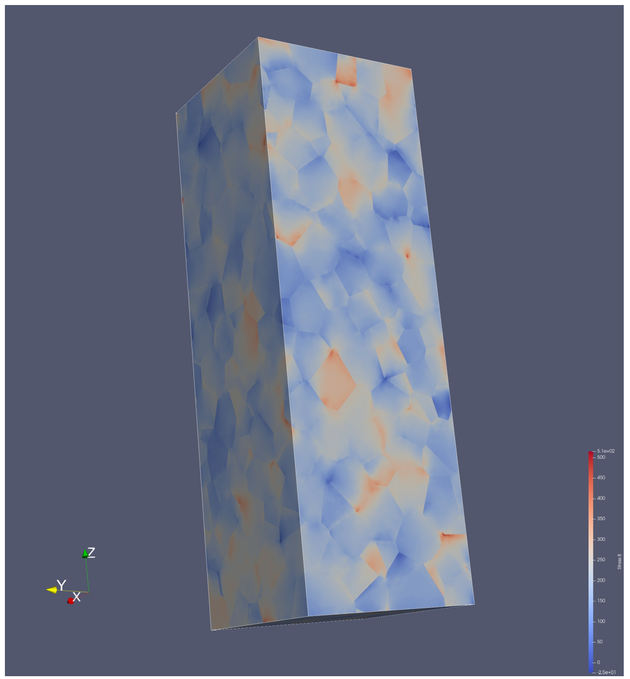
Mechanical Banding (MechBands)

Example 2: Comparison of Deformation Bands to Strength-to-Stiffness Metric

Deformation state: Extension in z-direction
Solutions generated by FEpX and MechMet.
Polycrystal has 622 grains.
Mesh has ~225,000 elements.



Grains with Mesh

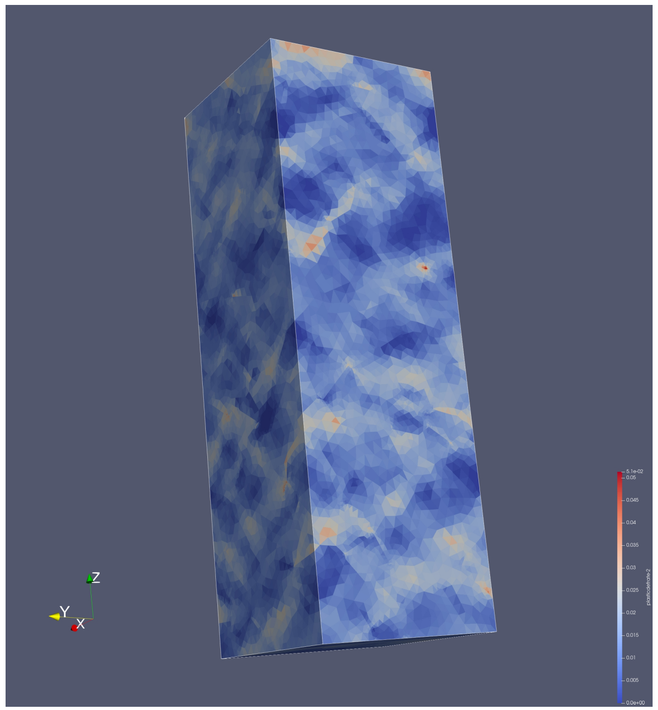


Axial Stress @ 0.001 nominal elastic strain (MechMet)

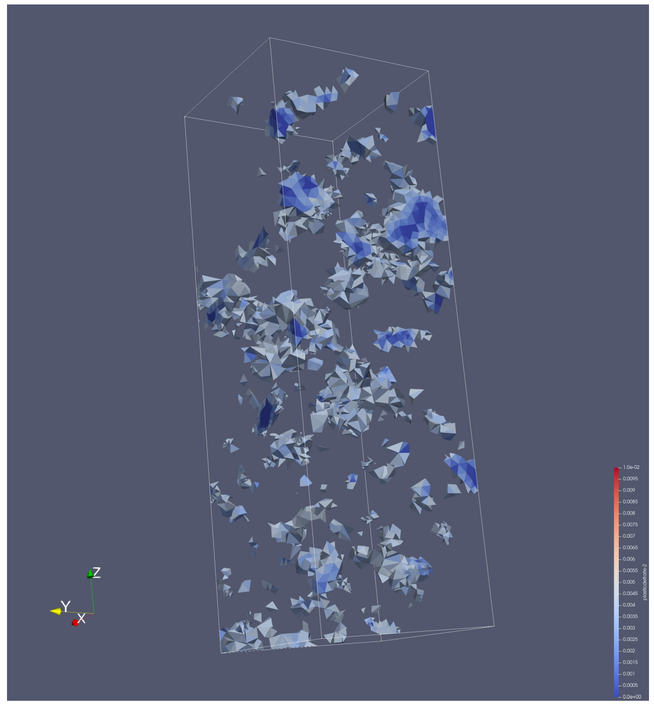
Mechanical Banding (MechBands)

Example 2: Comparison of Deformation Bands to Strength-to-Stiffness Metric

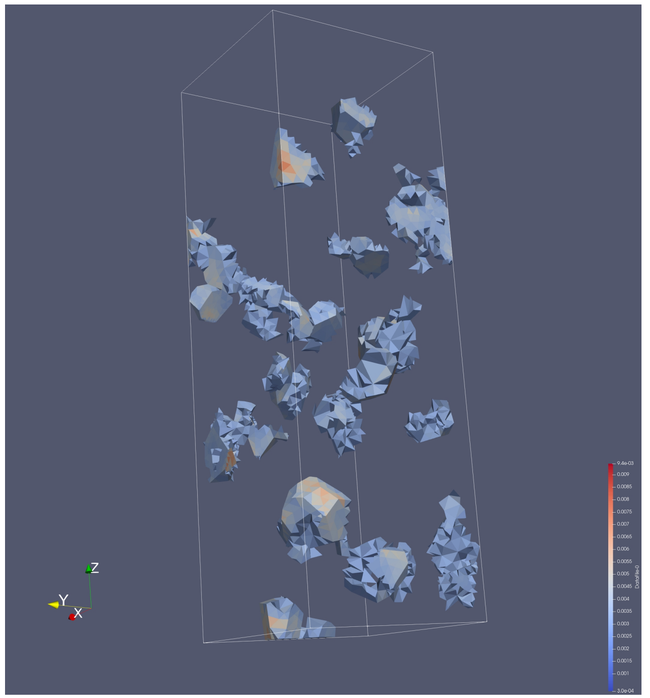
Effective plastic deformation rate near onset of yielding (FEpX)



Elements with low values of plastic deformation rate



Elements with high values of strength-to-stiffness



Mechanical Bands (MechBands)



Auxiliary Script: BigBandGraphs

Purpose:

BigBandGraphs processes the graph data generated by MechBands to create connected element lists. The connected elements all share values of the field variable within the specified range. The connected element lists offer the potential for correlating behaviors across the different field variables or across the same field variable at different points of a deformation history.

Methodology:

BigBandGraphs works with graph objects created in MechBands that define sets of connected elements that fall within specified upper and lower threshold values. One of the graphs generated in MechBands contains all of the graph links that connect mesh-based graph nodes (finite elements) that both fall within the threshold limits. While all graph links (element interfaces) connect two nodes of the graph, the links may not connect with others to form larger sets number of connected nodes. This means that the element sets may largely be disconnected. The graph routine in MatLat, conncomp, bins the elements into sets that are interconnected and identifies the degree of connectedness (number of elements) for each set. Using this information, BigBandGraphs ranks the sets by size, identifies the associated finite elements, and defines a new field variable equal to the set number. This field is used in Paraview to display the larger sets (bands). This capability is redundant to the capabilities of MechBands, but uses the graph theory routines available in MatLab. This opens the door to further analyses of connected set statistics available in MatLab.

BigBandGraphs second function is to create plots of the band graphs created in MechBands for the largest bands found using the MechBands search method. The plots display the topology of the network graph. These plots are messy and their utility is a bit unclear.

Results:

The MatLab-based routines based on graph analyses give identical bands as the MechBands search method, as shown in Slide 5.